ReDI School of Digital Integration

Data Circle 2025
Sprint 2

Project A:
Flu Shots

We use tech to connect human potential and opportunity with dignity & humility

Team "Halva" - Project A-
Flu Shot Learning: Predict H1N1 and Seasonal Flu Vaccines

Project Goal

Predict how likely individuals are to receive their H1N1 and seasonal flu vaccines.

Specifically, predicting two probabilities: one for **h1n1_vaccine** and one for **seasonal_vaccine.**

Each row in the dataset represents one person from USA who responded to the National 2009 H1N1 Flu Survey.

ReDI School of
Digital Integration

Objectives

- ❏ Optimizing machine learning pipeline
- ❏ Build predictive models for the h1n1 and seasonal vaccines uptake
- ❏ Evaluate multiple classification algorithms to identify the best-performing models for both vaccine targets
- ❏ Optimize model parameters to enhance predictive performance

We use tech to connect human potential and opportunity with dignity & humility

Baseline Model Creation

- ❑ Split data with stratified sampling to maintain class distribution
- ❑ Multiple classifiers are trained and compared:
  - Logistic Regression
  - Random Forest
  - XGBoost
  - LightGBM
- ❑ Each model is trained separately for the two vaccine targets.
- ❑ Additional set of models which are trained for both vaccines.

We use tech to connect human potential and opportunity with dignity & humility

ReDI School of Digital Integration

Model Selection and Comparison before balancing the data

- ❏ Compare models on performance metrics
- ❏ Analyze how model performance differs across vaccine targets

Summary Report:

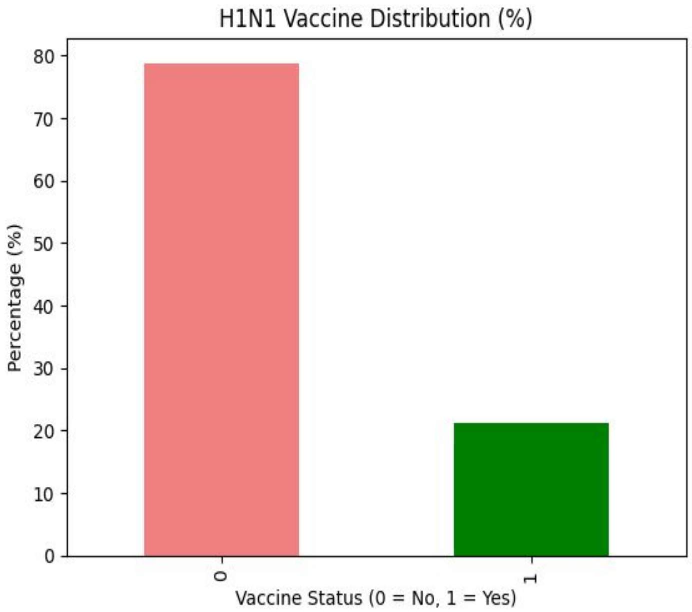
	Validation Accuracy		f1	
Target	h1n1_vaccine	seasonal_vaccine	h1n1_vaccine	seasonal_vaccine
Model				
LightGBM	0.839386	0.785474	0.559096	0.765452
Logistic Regression	0.814302	0.754399	0.405276	0.728701
Random Forest	0.831898	0.782666	0.488027	0.759278
XGBoost	0.832273	0.769936	0.546099	0.747483

We use tech to connect human potential and opportunity with dignity & humility

ReDI School of Digital Integration

Dealing with imbalancement of the h1n1 vaccine

- ❑ Without Balancing: F1-score for the H1N1 is 55.4
- ❑ Strategy 1. Random Oversampling: F1-score for the H1N1 is 59.8
- ❑ Strategy 2. Random Undersampling: F1-score for the H1N1 is 59.7
- ❑ Strategy 3. Oversampling with SMOTE: F1-score for the H1N1 is 59.8
- ❑ Strategy 4. Label Weight Balancing for both vaccines: F1-score for the H1N1 is 59.7



We use tech to connect human potential and opportunity with dignity & humility

ReDI School of Digital Integration

Model Selection and Comparison after balancing the data

- ❑ Compare models on performance metrics
- ❑ Analyze how model performance differs across vaccine targets
- ❑ Stratified k-fold cross-validation

=== Summary Report ===

Metrics reported: Validation Accuracy, F1 Score, ROC AUC, Cross-Validation Means/Stds, ROC Curve Image Path

Model	Target	Validation Accuracy	F1 Score	AUC	CV Accuracy Mean	CV Accuracy Std	CV F1 Mean	CV F1 Std	CV AUC Mean	CV AUC Std
Logistic Regression	h1n1_vaccine	0.747473	0.546250	0.801328	0.744000	0.003234	0.547551	0.003749	0.811679	0.004173
Random Forest	h1n1_vaccine	0.834706	0.508625	0.820970	0.835923	0.002897	0.547503	0.010136	0.836058	0.005671
XGBoost	h1n1_vaccine	0.837514	0.551653	0.835423	0.840604	0.003611	0.553708	0.012655	0.842980	0.004393
LightGBM	h1n1_vaccine	0.836765	0.551440	0.832714	0.809076	0.002696	0.598543	0.003512	0.833651	0.004457
Logistic Regression	seasonal_vaccine	0.772183	0.756746	0.844602	0.769648	0.005999	0.756526	0.006793	0.842957	0.003757
Random Forest	seasonal_vaccine	0.785661	0.764839	0.856031	0.781780	0.004055	0.762437	0.003699	0.855282	0.003951
XGBoost	seasonal_vaccine	0.789218	0.767931	0.862201	0.786723	0.006273	0.766933	0.006028	0.861134	0.002977
LightGBM	seasonal_vaccine	0.784350	0.769692	0.861201	0.783802	0.005348	0.771317	0.004037	0.860252	0.002822

Hyperparameter Tuning

- ❑ Define search spaces for key hyperparameters in top models.
- ❑ Use grid or random search with cross-validation focused on ROC AUC scoring.
- ❑ Evaluate tuned models against baselines to check improvements and overfitting risk.

Vaccine	H1N1	Seasonal vaccine
Model	LightGBM	LightGBM
Learning rate	0.1	0.1
N-estimators	200	100
Number of leaves	63	31

We use tech to connect human potential and opportunity with dignity & humility

Final Training and Prediction

- ❏ Models are retrained on the full training set.
- ❏ Predictions (probabilities) are generated for the test set.
- ❏ Generated visualization of model performance metrics

ReDI School of Digital Integration

Challenges

- Imbalanced data for h1n1 vaccine
- Debugging and optimizing the preprocessing pipeline
- Figuring out whether the model for the H1N1 is performing well or not

We use tech to connect human potential and opportunity with dignity & humility

Conclusion

- LightGBM is the best performing model for both of the targets, based on the cross-validation and F1-score.

Thank you!

We use tech to connect human potential and opportunity with dignity & humility